

DEPARTMENT OF MINES

SOUTH AUSTRALIA

MAGNETIC OBSERVATIONS IN THE MIDDLEBACK RANGES, 1950

GEOPHYSICAL REPORT 1/51

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SUMMARY.

Magnetic observations have been used to trace the continuity of hematite-quartzite beds under shallow sand or alluvial cover in the vicinity of the Middleback Ranges. In some cases continuity of the beds occurs under the alluvial cover in other cases the beds appear to be discontinuous.

INTRODUCTION.

It was demonstrated by C. Kerr Grant (Mining Review No. 90), that the steeply dipping hematite quartzite beds in this area, produced anomalies in the vertical component of the magnetic intensity of the order of several thousand gammas.

In this survey an attempt has been made to utilise the above property to demonstrate the subsurface continuity, or otherwise, of the iron quartzite beds, where no surface outcrops can be found.

METHOD.

As no general survey of this area, which is mainly of a slight undulating nature, covered by mallee etc, was available, traverses were established from points whose relative positions were fixed by the use of aerial photographs.

Readings were taken with a Hilger and Watts vertical force vatiometer, having a scale factor of approximately 30 gammas per division.

RESULTS.

The results are shown in the form of traverses plotted on the preliminary regional geological map prepared by Dr. K.R. Miles.

Over the hematite quartzite formations anomalies of up to 8,000 gammas were recorded.

Over the beds of the Camel Range and its extensions, where the iron content is largely in the form of magnetite or magnetic hematite, high readings were obtained, due mainly, as is indicated by the width of the anomalies, to the unoxidised magnetite at a depth two to three hundred feet below the surface.

This type of formation may be traced by this method, and it is expected that the traverses taken through areas where no geological

outcrops have been found will prove a useful addition to the geological information.

As may be seen from the plotted traverses, some of these areas of no outcrop are due to erosion of the beds until no exposures have been left, although the magnetic anomaly is still in evidence.

In some areas there is no magnetic anomaly, where from the exposed geological evidence it could be reasonably assumed that the beds should be continuous under the surface cover.

It is anticipated that further comment on this situation will be provided by Dr. Miles in his reports to be published on completion of his work in this area.

Over the Ash Range, where the beds consist of a limonite chert, the magnetic properties of which are not far different from the surrounding rocks, little or no change is apparent in the magnetic values, and the method appears to be of no value in tracing this type of formation.

South of the Middleback Range along the main Port Lincoln Highway, there were a number of anomalous readings, the ones of greater magnitude appearing to be associated with the contacts between various rock types, e.g. the most pronounced anomaly is in the area where there is a transition from the conglomerates etc. of the Moonabee Range formation to the porphyry and granite area further west.

This anomaly appears to persist in the readings taken along the most southerly traverse to date, where much of the country is covered by sand dunes etc. and may be of use in tracing the contact area in this type of country.

Immediately south of the Middleback Range, there is a region in which the value of the vertical magnetic intensity is low indicating that the hematite quartzite formation of the Main Middleback Range does not cross the road, but ends north of it.

Plan Reference -

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L48/27 } MISSING

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